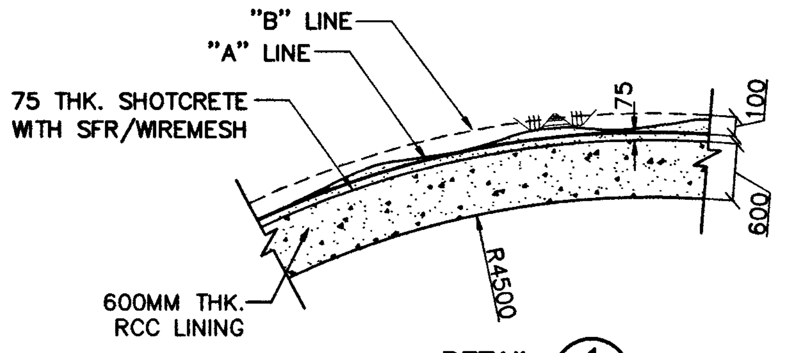
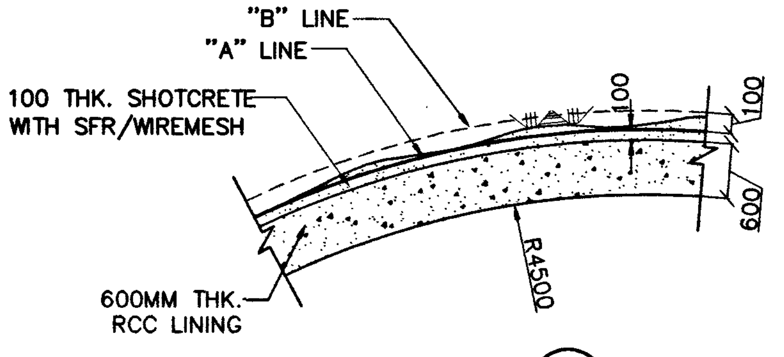
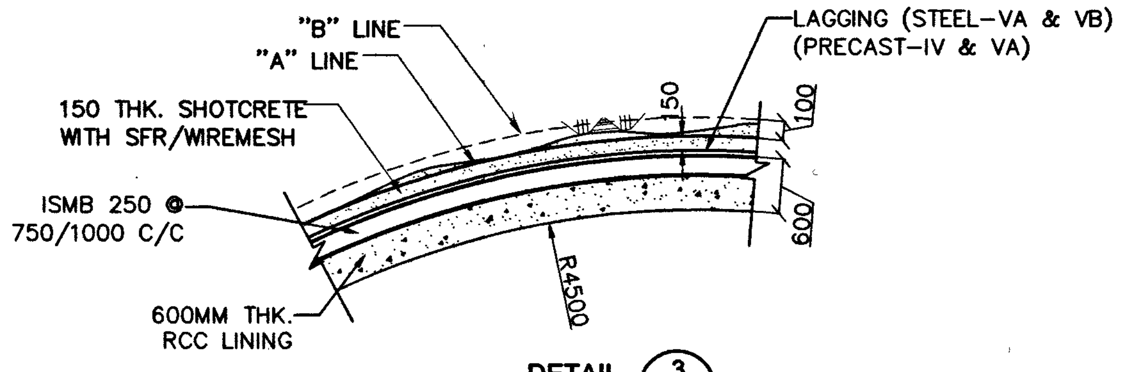


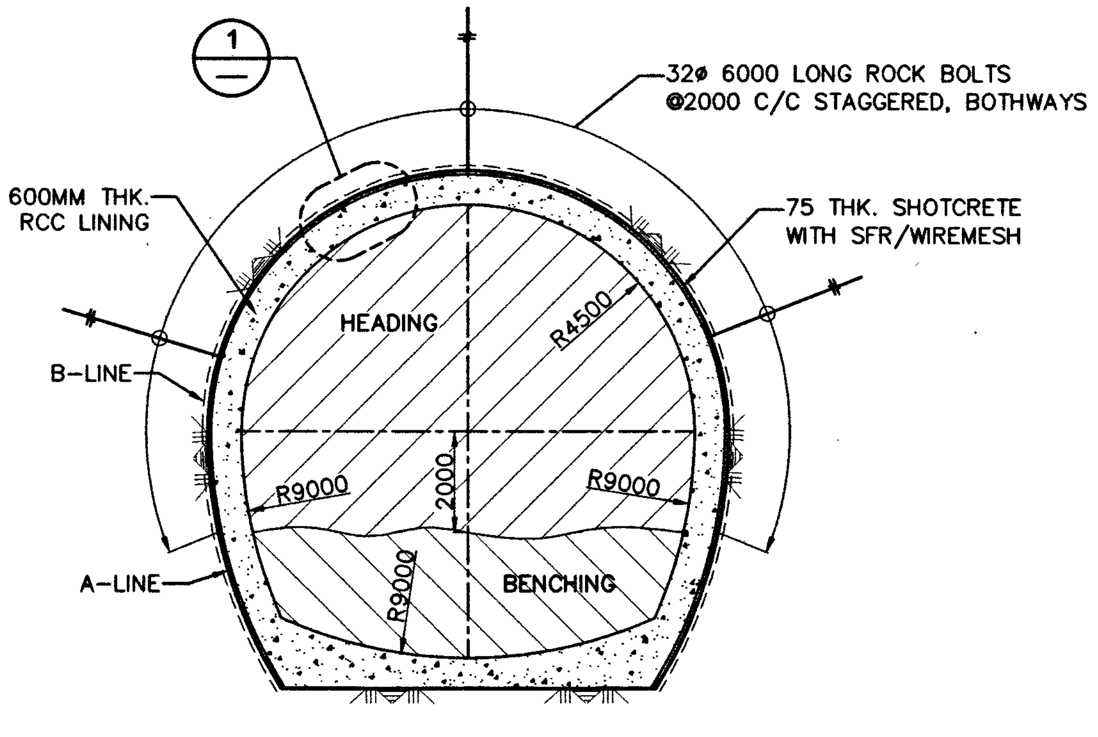
DETAIL ① (NTS)



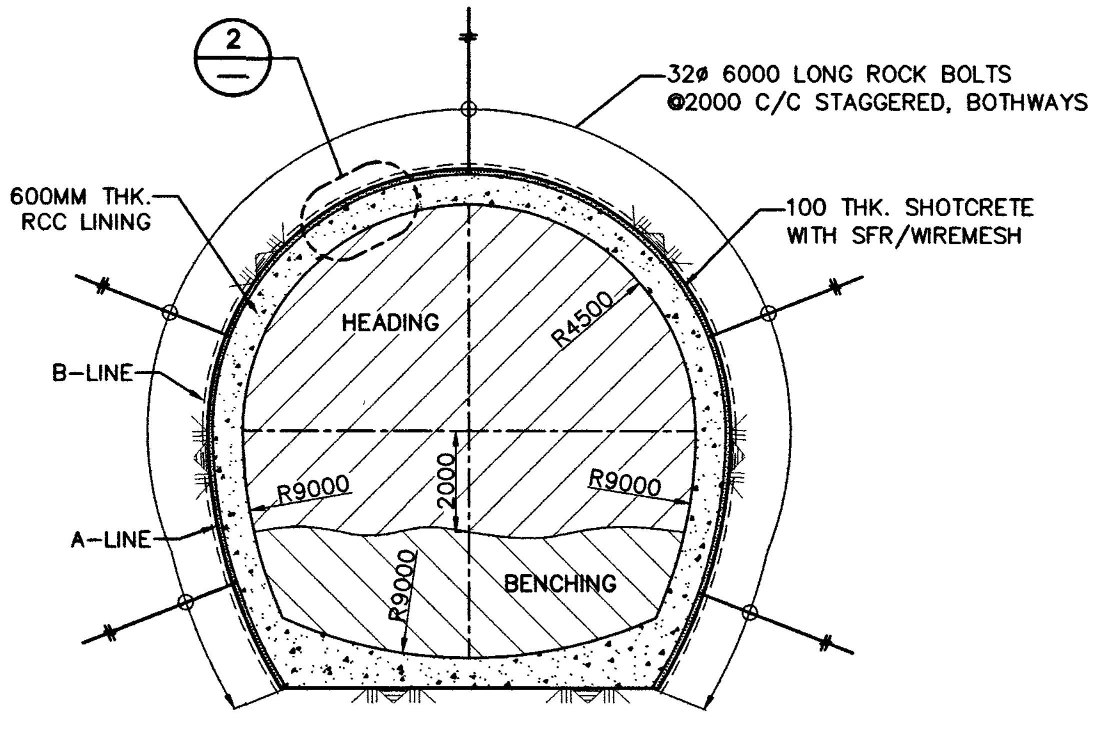
DETAIL ② (NTS)


"B" LINE
"A" LINE
LAGGING (STEEL-VA & VB) (PRECAST-IV & VA)
150 THK. SHOTCRETE WITH SFR/WIREMESH
ISMB 250 @ 750/1000 C/C
150
100
600
R4500
600MM THK. RCC LINING

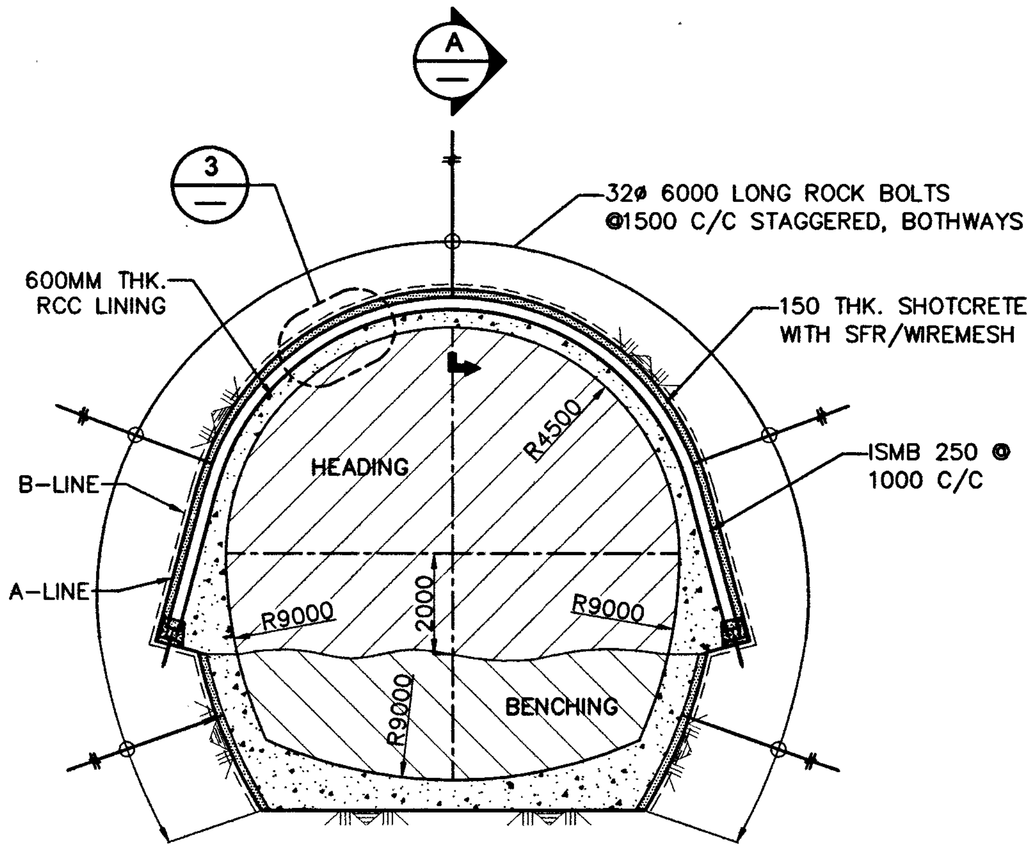
DETAIL ③ (NTS)


①
32ø 6000 LONG ROCK BOLTS @2000 C/C STAGGERED, BOTHWAYS
600MM THK. RCC LINING
75 THK. SHOTCRETE WITH SFR/WIREMESH
HEADING
R4500
B-LINE
R9000
2000
R9000
A-LINE
R9000
BENCHING

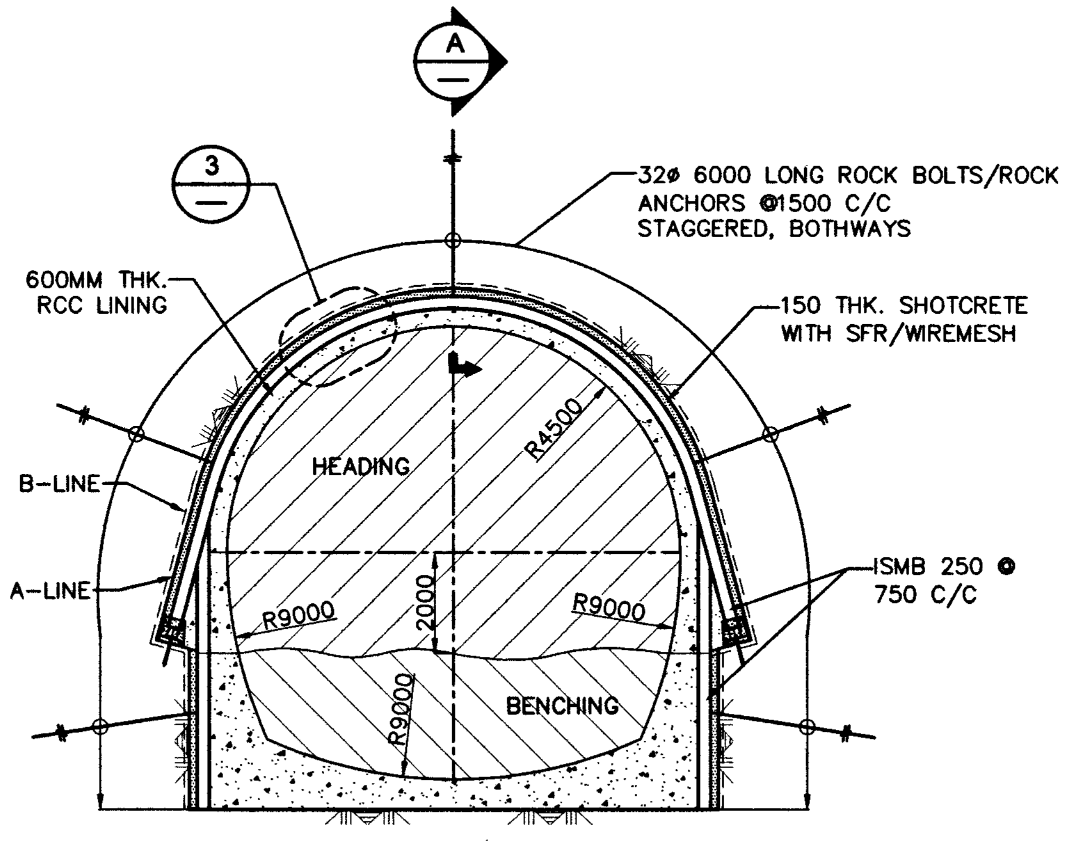
ROCK SUPPORT (CLASS-II)


②
32ø 6000 LONG ROCK BOLTS @2000 C/C STAGGERED, BOTHWAYS
600MM THK. RCC LINING
100 THK. SHOTCRETE WITH SFR/WIREMESH
HEADING
R4500
B-LINE
R9000
2000
R9000
A-LINE
R9000
BENCHING

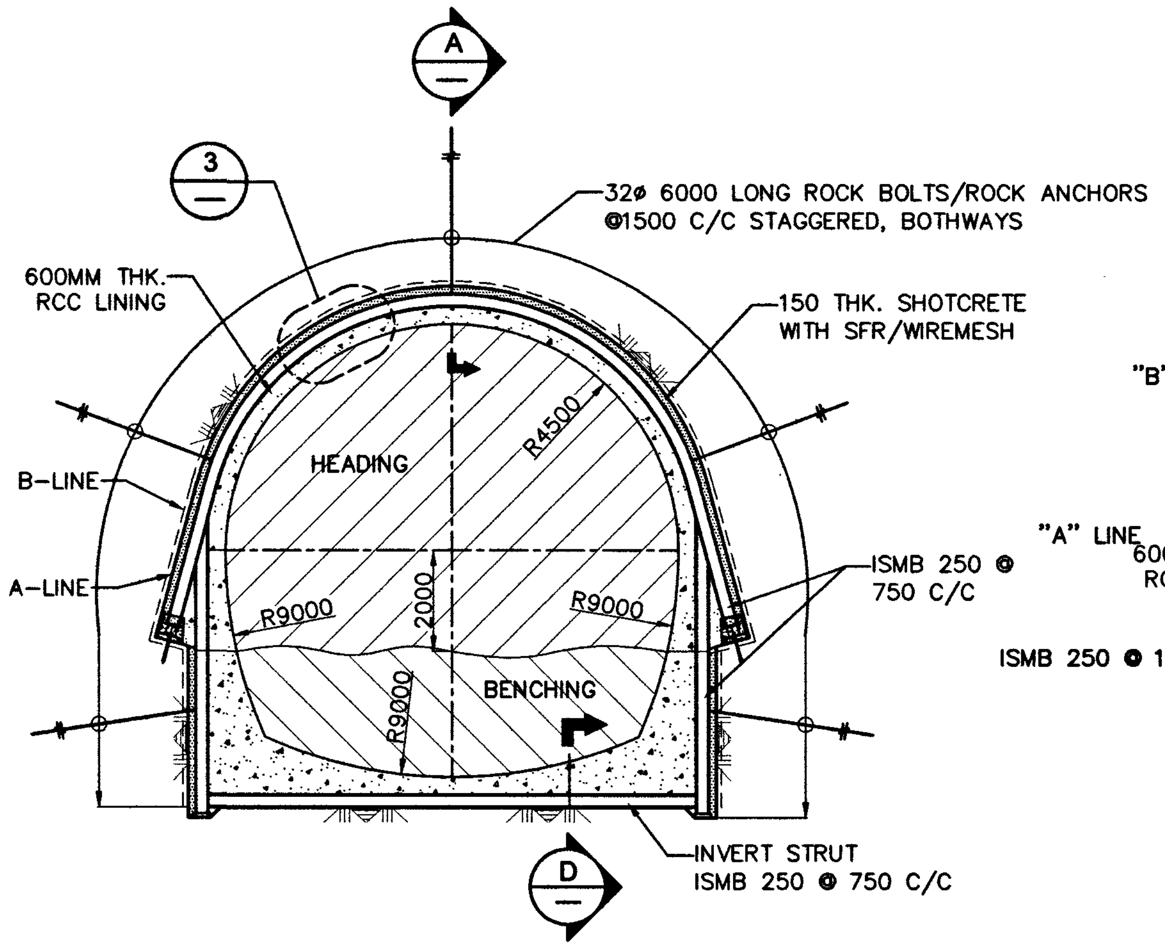
ROCK SUPPORT (CLASS-III)



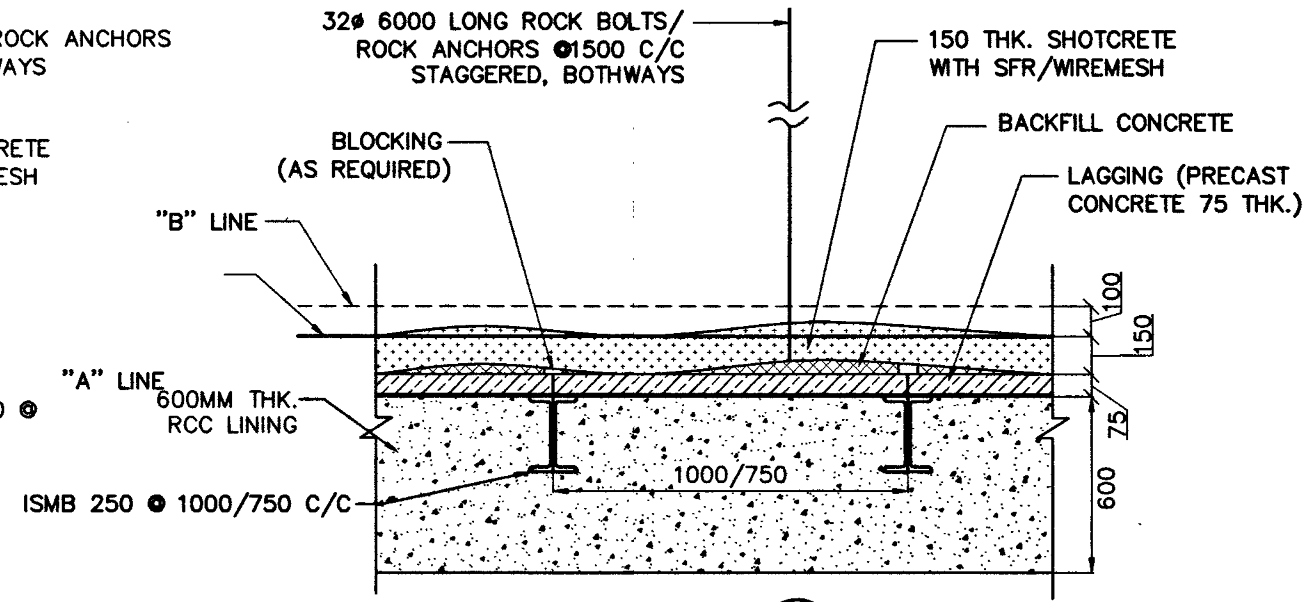
ROCK SUPPORT (CLASS-IV)


A
③
32ø 6000 LONG ROCK BOLTS/ROCK ANCHORS @1500 C/C STAGGERED, BOTHWAYS
600MM THK. RCC LINING
150 THK. SHOTCRETE WITH SFR/WIREMESH
HEADING
R4500
B-LINE
A-LINE
ISMB 250 @ 750 C/C
R9000
2000
R9000
R9000
BENCHING

ROCK SUPPORT (CLASS-VA)

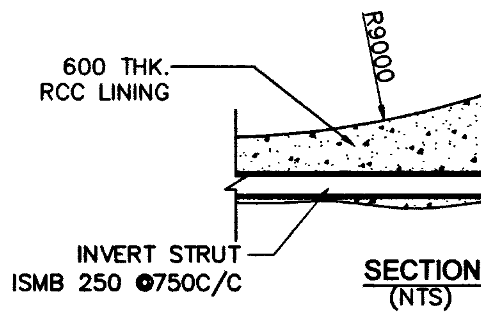

A
③
32ø 6000 LONG ROCK BOLTS/ROCK ANCHORS @1500 C/C STAGGERED, BOTHWAYS
600MM THK. RCC LINING
150 THK. SHOTCRETE WITH SFR/WIREMESH
HEADING
R4500
B-LINE
A-LINE
R9000
2000
R9000
ISMB 250 @ 750 C/C
R9000
BENCHING
D
INVERT STRUT ISMB 250 @ 750 C/C

ROCK SUPPORT (CLASS-VB)

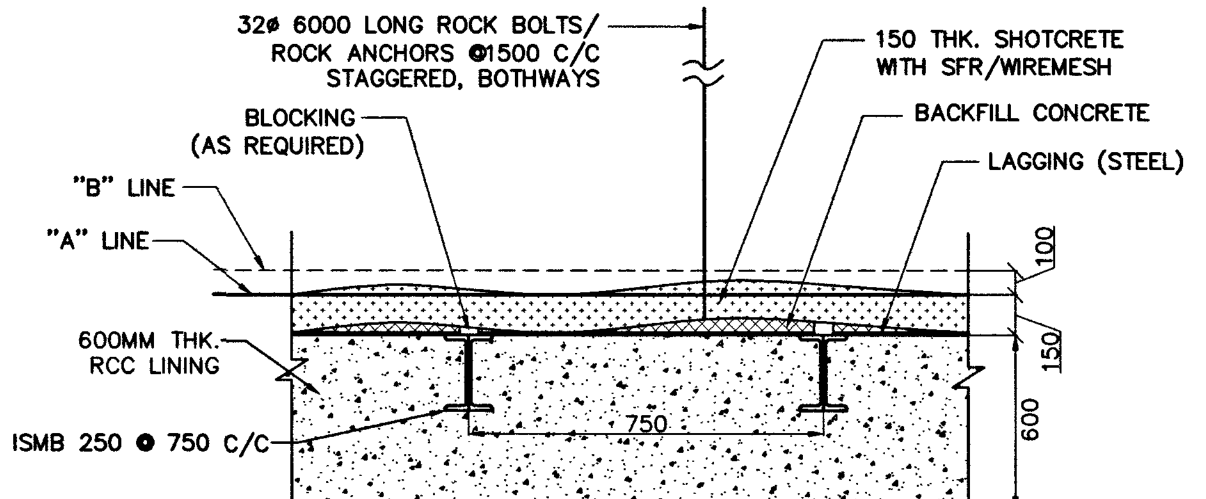

32ø 6000 LONG ROCK BOLTS/ ROCK ANCHORS @1500 C/C STAGGERED, BOTHWAYS
150 THK. SHOTCRETE WITH SFR/WIREMESH
BACKFILL CONCRETE
BLOCKING (AS REQUIRED)
LAGGING (PRECAST CONCRETE 75 THK.)
"B" LINE
"A" LINE
600MM THK. RCC LINING
100
150
75
600
1000/750
ISMB 250 @ 1000/750 C/C

SECTION Ⓐ (NTS)

(CLASS-IV & VA - WITH CONCRETE LAGGING)

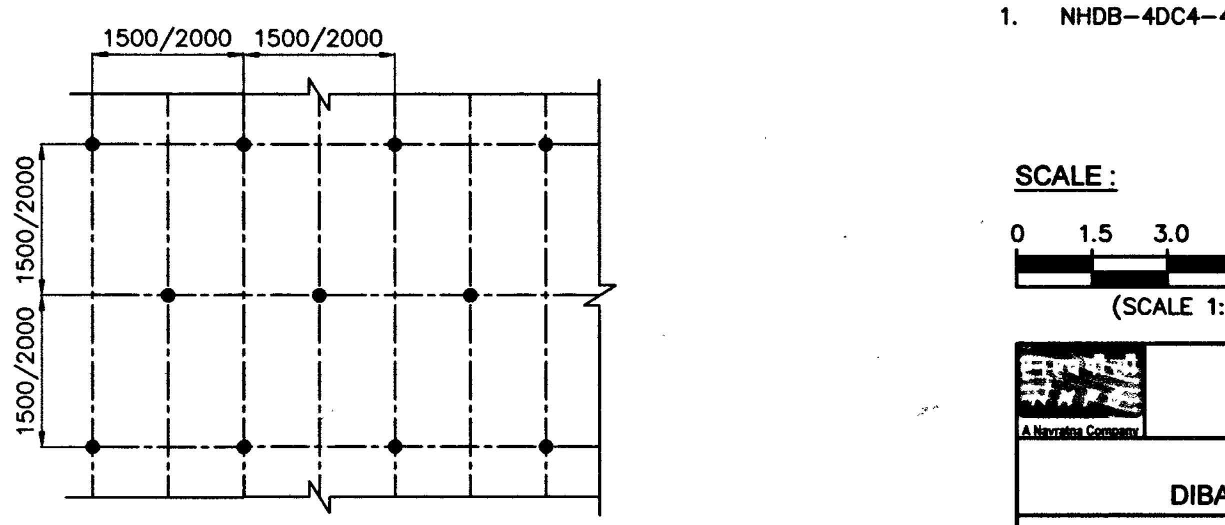


SECTION Ⓓ (NTS)



SECTION Ⓐ (NTS)

(CLASS-VA & VB - WITH STEEL LAGGING)



TYPICAL ROCK BOLT/ROCK ANCHORS PATTERN
(NTS)

NOTES:-

1. सभी माप मिलीमीटर में तथा ऊंचाई मीटर में दिए गए हैं। ALL DIMENSIONS ARE IN MILLIMETERS AND LEVELS IN METERS.
2. THE DEFINITION OF 'A' , 'B' LINE AND VARIOUS CLASSES OF EXCAVATION SHOWN SHALL BE AS PER SECTION B.3 OF TECHNICAL SPECIFICATIONS.
3. UNDERGROUND EXCAVATION INCLUDING PROTECTION OF TUNNEL INVERT SHALL BE CONFORMING TO SECTION B.3 OF TECHNICAL SPECIFICATION.
4. ROCK BOLTS/ANCHORS SHOWN IN THE DRAWING SHALL BE CONFORMING TO SECTION B.4 OF TECHNICAL SPECIFICATION.
5. MINOR MODIFICATION IN THE LENGTH, ORIENTATION,PATTERN AND LOCATION OF ROCK BOLTS/ANCHORS MAY BE DONE TO SUIT THE SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
6. STEEL RIBS SHOWN IN THE DRAWING SHALL BE CONFORMING TO SECTION B.4 OF TECHNICAL SPECIFICATION. THE SPACING OF RIBS SHOWN MAY BE MODIFIED TO SUIT THE SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN- CHARGE.
7. STEEL LAGGING OR PRE CAST CONCRETE LAGGING AS SPECIFIED IN SECTION B.4 OF TECHNICAL SPECIFICATION SHALL BE USED WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
8. IN CLASS-VB TYPE OF ROCK/SQUEEZING ZONES, INVERT STRUT MAY ALSO BE REQUIRED AS PER SITE SPECIFIC CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
9. SFR SHOTCRETE SHOWN IN THE DRAWING SHALL BE APPLIED AS SPECIFIED IN SECTION B.5 OF TECHNICAL SPECIFICATION. IF REQUIRED PLAIN SHOTCRETE WITH WIREMESH CAN BE USED WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
10. 45ø 5000 LONG (INTO ROCK) CONSOLIDATION GROUTING @ 3000 C/C STAGGERED BOTHWAYS SHALL BE PROVIDED IN CLASS IV, VA & VB OR AS DIRECTED BY ENGINEER-IN-CHARGE, AS PER SECTION B.7 OF TECHNICAL SPECIFICATION.
11. FOR ADDITIONAL ROCK SUPPORT MEASURES IN POOR ROCK/REACHES REFER DRG. NO. NHDB-4BC4-41-DD-1007.
12. PREGROUTING/FOREPOLING/PIPE ROOFING MAY BE USED AS PER SITE SPECIFIC CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
13. M25/A40 GRADE FOR INVERT/OVERT CONCRETE CONFORMING TO SECTION B.9 OF TECHNICAL SPECIFICATION SHALL BE USED.
14. THIS DRAWING IS ONLY VALID FOR TAIL RACE TUNNEL.
15. FOR STEEL RIBS DETAILS REFER DRG. NO. NHDB-4DC4-41-DD-1010 & 1011.

REFERENCE DRAWINGS :-

1. NHDB-4DC4-41-DD-1010 & 1011

SCALE :

0 1.5 3.0 4.5 6.0 7.5 METRE

(SCALE 1:150)

एन एच पी सी लिमिटेड
(भारत सरकार का एक नवरत्न उद्यम)
NHPC Limited
(A GOVT. OF INDIA NAVRATNA ENTERPRISE)
A Navratna Company

दिबांग बहुउद्देशीय परियोजना
DIBANG MULTIPURPOSE PROJECT

TAIL RACE TUNNEL
ROCK SUPPORT & CONCRETE LINING DETAILS

DATE	NO.	REVISION OR ISSUES	BY	CH.	APP
26.09.25	00	ISSUED FOR CONSTRUCTION			

DESIGN MEMO NO. डिजाइन मेमो संख्या

DRAWN उद्रेखित ABHISHEK YADAV
SUBMITTED प्रस्तुत ASHISH RAO
RECOMMENDED अनुशंसित SAURABH SINGH
APPROVED अनुमोदन S CHOWDHURY

DATE दिनांक SEPT 2025
DRG. NO. उद्रेख संख्या NHDB 4DC4 41 DD 1009 00

IMS/DED/F-02 REV. NO. : 03 EFF. DATE 21.07.2016

DRG. NO. NHDB-4DC4-41-DD-1009-00

INKJET 75 MIC D/W BH.